ENME 351: Electronics and Instrumentation II

Lab 6: Digital to Analog Conversion and Servo Motors

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Overview

In this lab, a servo motor was controlled using two different methods:

1. A potentiometer for continuous input control
2. A 3-bit R–2R DAC for discrete digital input control

The DAC output voltage was read by the Arduino and mapped into a PWM signal to position the servo motor.

Exercise 1: 3-Bit R–2R DAC Theory

Problem 1

Determine the theoretical output voltages of a 3-bit R–2R DAC with $V_{ref} = 5$ V, and determine the least significant bit voltage.

Solution 1

For a 3-bit DAC, the analog output is given by

$$V_{out} = V_{ref}\left(\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right),$$

where b_2, b_1, and b_0 are the bit values of the input.
Since

$$V_{ref} = 5 \text{ V},$$

the least significant bit voltage is

$$V_{LSB} = \frac{5}{2^3} = \frac{5}{8} = 0.625 \text{ V}.$$

Thus the eight possible output voltages are:

Bits	Decimal Value	Voltage (V)
000	0	0.000
001	1	0.625
010	2	1.250
011	3	1.875
100	4	2.500
101	5	3.125
110	6	3.750
111	7	4.375

$$V_{LSB} = 0.625 \text{ V}$$

Exercise 2: Circuit Simulation

Problem 2

Simulate the 3-bit R–2R DAC in CircuitLab and verify the output voltages for selected bit settings.

Solution 2

The DAC was simulated in CircuitLab using resistor values:

$$R = 10\ \text{k}\Omega, \qquad 2R = 20\ \text{k}\Omega.$$

Two example simulated input cases are shown below.

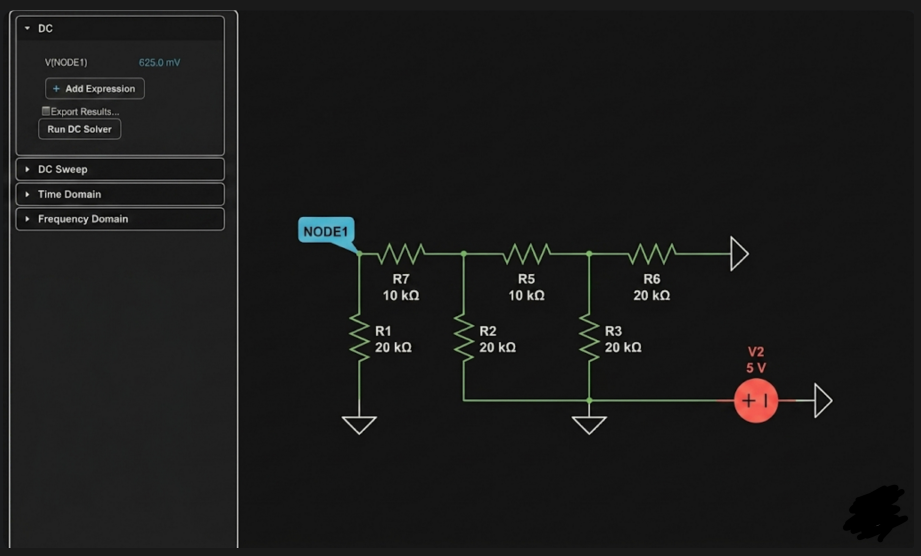

Figure 1: CircuitLab simulation for input 001.

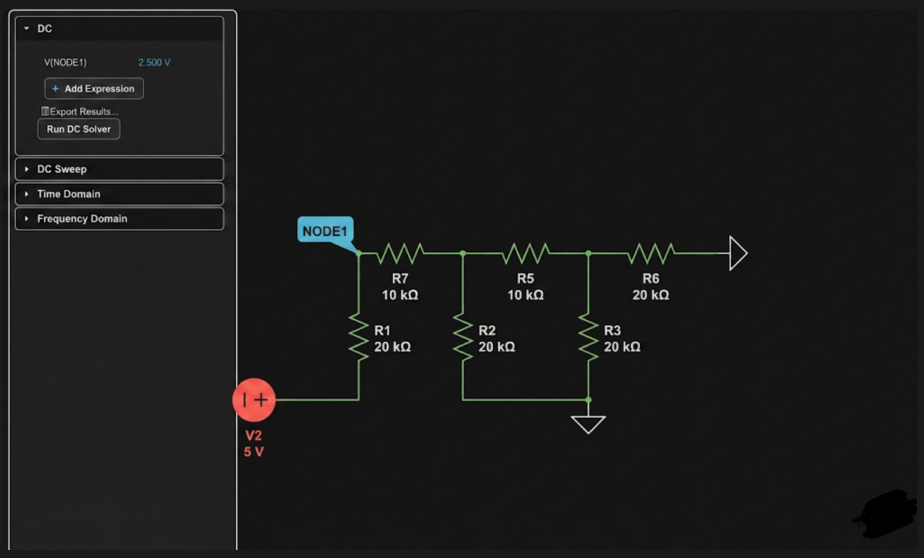

Figure 2: CircuitLab simulation for input 100.

These simulation results were used to confirm that the DAC output followed the expected binary-weighted voltage levels.

Exercise 3: Physical DAC Construction

Problem 3

Build the 3-bit R–2R DAC physically and compare measured voltages to theoretical predictions.

Solution 3

The physical DAC was built on a breadboard using:

$$R = 10\ \text{k}\Omega, \qquad 2R = 20\ \text{k}\Omega.$$

Each bit input was manually connected to either:

- 5 V for logic 1
- ground for logic 0

The output voltage was measured using the Arduino analog input.

> Small differences between theoretical and measured values are expected due to resistor tolerances, breadboard parasitics, and ADC quantization error.

Exercise 4: Servo Control Using DAC

Problem 4

Use the DAC output voltage to control the position of the servo motor and discuss the number of possible positions.

Solution 4

The DAC output was connected to Arduino analog pin $A0$. The Arduino read the analog voltage and converted it into a PWM pulse width for the servo motor.

Solution 4

The signal chain was:

Digital bits → DAC output voltage → Arduino ADC → PWM signal → Servo position

Because the DAC is 3-bit, the number of possible input combinations is

$$2^3 = 8.$$

Therefore, the servo motor can achieve **8 discrete positions**.
Unlike the potentiometer-based control, which allowed nearly continuous motion, the DAC-based control produced stepped motion because only eight voltage levels were available.

A 3-bit DAC provides 8 discrete voltage levels, so the servo can only move to 8 distinct angular positions. Increasing the number of DAC bits would produce smaller voltage steps and finer angular control.

Discussion

Comparison of control methods

Method	Resolution	Behavior
Potentiometer	Continuous / high resolution	Smooth motion
3-bit DAC	8 levels	Stepped motion

How to improve the DAC

- Use more bits, such as a 4-bit or 8-bit DAC
- Use precision resistors for better accuracy
- Use a dedicated DAC integrated circuit

Conclusion

Solution
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This lab demonstrated the design, simulation, and implementation of a 3-bit R–2R DAC and its application to servo motor control.
The potentiometer provided smooth analog control, while the DAC provided discrete step-based control. The lab also illustrated a complete signal path from digital input to analog conversion, analog measurement, PWM generation, and mechanical output.